# Frontiers of Network Science 4250
# Assignment 1: Network Analysis and Visualization

**Due: 11:59 PM on Friday, October 23, 2026**

## Overview

You will analyze the two real-world networks approved during the Network Selection stage and compare them with corresponding Erdős-Rényi (ER) random graphs and Barabási-Albert (BA) scale-free networks. You will work with six networks in total: two real-world networks and, for each one, an ER and a BA counterpart.

## Network Generation

For each approved real-world network, create:

- an Erdős-Rényi (ER) random graph with the same number of nodes and approximately the same number of edges; and
- a Barabási-Albert (BA) scale-free network with the same number of nodes and approximately the same number of edges.

**ER:** Use the average degree of the corresponding real network to choose parameters that produce approximately the same number of edges.

**BA:** Choose the model parameter so that the generated network has an edge count as close as possible to that of the real network.

## Common Analysis Rules

- Disconnected networks: compute shortest-path-based measurements on the largest connected component (LCC). For directed networks, apply this rule using the largest weakly connected component.
- Component statistics: compute the number of components and component-size distributions on the entire network, not only on the LCC.
- Preprocessing: document any transformation of the original network; if graph size changes, report the final number of nodes and edges used in the analysis.
- Consistency: within each real/ER/BA set, use consistent analysis conventions whenever possible.

## Task 1 - Real-World Network Description (5 points)

For each approved real-world network, describe the original dataset. Include:

- network name and data source;
- what the nodes and edges represent;
- how, when, and by whom the network was collected or constructed, when available;
- number of nodes and edges in the original dataset;
- whether the network is directed/undirected and weighted/unweighted; and
- the significance, meaning, or context of the network.

*Your description should contain more information than the brief description submitted during Network Selection.*

| Criterion | Points |
|---|---|
| Clear explanation of what the nodes and edges represent | 2 |
| Dataset source, collection background, and context | 1 |
| Correct network size and structural metadata | 1 |
| Significance and overall completeness | 1 |
| **Total** | **5** |

## Task 2 - Global Network Properties (15 points)

For each of the six networks, compute and report:

**(i) Distance-based properties:** diameter, radius, and average eccentricity. For disconnected networks, use the LCC defined above and report its size.

**(ii) Connectivity:** number of connected components and size of the largest component. For directed networks, report weakly and strongly connected components and the size of the largest component of each type. Use the entire network.

**(iii) Degree statistics:** average node degree and degree variance. For directed networks, also report average in-degree and average out-degree where appropriate.

*Organize results so that the real, ER, and BA networks in each set can be compared directly.*

| Criterion | Points |
|---|---|
| Diameter, radius, and average eccentricity | 5 |
| Connected-component statistics | 5 |
| Average degree and degree variance | 3 |
| Correct handling/documentation of disconnected and directed networks | 2 |
| **Total** | **15** |

## Task 3 - Network Distributions (15 points)

For each of the six networks, compute and plot:

**(iv) Degree distribution.** For directed networks, also plot in-degree and out-degree distributions.

**(v) Distribution of each node's longest shortest-path distance.** For each node in the analyzed component, determine the length of its longest shortest path to another reachable node. For disconnected networks, use the LCC defined above.

**(vi) Clustering coefficient distribution.**

**(vii) Connected-component size distribution.** Use the entire network.

For measurements (v)-(vii), also report (viii) the average value and (ix) the variance.

**Plot requirements:** Use a logarithmic vertical axis or log-log scale when appropriate. Include an appropriate best-fit line on each distribution plot. Use informative titles, labeled axes, and legends when needed.

**Hint:** For Task 2(i) and Task 3(v), repeatedly use breadth-first search/walk from individual nodes to determine shortest-path distances and each node's longest shortest-path distance.

| Criterion | Points |
|---|---|
| Degree distributions | 4 |
| Longest-shortest-path distance distribution | 3 |
| Clustering coefficient distribution | 3 |
| Connected-component size distribution | 2 |
| Means and variances for (v)-(vii) | 1 |
| Plot quality, labeling, scales, and best-fit lines | 2 |
| **Total** | **15** |

## Task 4 - Comparison and Interpretation (9 points)

For each real/ER/BA set, compare the structural characteristics of the three networks using the results from Tasks 2 and 3. Go beyond stating that values or plots are different: explain what the differences indicate about the real-world network and how well the ER and BA models reproduce its properties.

Where relevant, discuss degree heterogeneity, connectivity/component structure, distance and path-length behavior, clustering, and other notable differences. Identify which real-world properties are reasonably captured by ER or BA and which are not.

| Criterion | Points |
|---|---|
| Comparison of the real networks with their corresponding ER networks | 2 |
| Comparison of the real networks with their corresponding BA networks | 2 |
| Use of quantitative results and plots from Tasks 2 and 3 as supporting evidence | 2 |
| Interpretation of which real-world properties are or are not captured by ER/BA | 2 |
| Overall clarity, depth, and coherence | 1 |
| **Total** | **9** |

## Task 5 - Methodology and Reproducibility (6 points)

Document the workflow so that another person could reproduce your analysis from the original datasets.

**Preprocessing:** describe, when applicable, removal of self-loops/duplicate edges, handling of missing data or edge weights, directed/undirected conversion, filtering, and component extraction. If preprocessing changes graph size, report the final analyzed network size and explain the difference from Task 1.

**ER/BA generation:** report number of nodes, relevant model parameter(s) (e.g., p for ER and m for BA), resulting number of edges, and random seed if used.

**Tools and files:** report tool/software name, version when available, tasks for which it was used, and important settings. Submit all relevant source code, scripts, notebooks, project files, or other supporting files, with sufficient run instructions.

| Criterion | Points |
|---|---|
| Preprocessing and final analyzed networks clearly documented | 2 |
| ER/BA generation parameters and resulting graph sizes documented | 1 |
| Tools, versions, and important settings identified | 1 |
| Source code/scripts/supporting files included | 1 |
| Workflow and instructions sufficient for reproduction | 1 |
| **Total** | **6** |

## Optional Task - Network Visualization (up to +10 bonus points)

Create a high-quality vector graphic representation of each real-world network. If a network is too large to display clearly, use and explain filtering, aggregation/collapsing, hierarchical representation, selective labeling, or other clutter-reduction techniques. Use meaningful visual encodings (e.g., node color/size or edge width), include a legend, justify major design choices, and export in SVG or PDF when possible.

| Criterion | Points |
|---|---|
| Visualization of Real Network 1 | 2 |
| Visualization of Real Network 2 | 2 |
| Appropriate layout and reduction of visual clutter | 2 |
| Meaningful use of visual encodings | 2 |
| Legends, explanation, and overall presentation quality | 2 |
| **Total** | **10** |

## Submission

Submit through Submitty by the deadline shown above. Include:

- a PDF report containing all required answers, tables, plots, comparisons, and any optional visualizations;
- source code/scripts/notebooks/project files and supporting files needed to reproduce the results; and
- brief run instructions if they are not already included in the report.

Package supplementary files in a single ZIP archive if necessary. Make sure submitted files are readable and dependencies are documented.

## Grading and Academic Integrity

**Required total: 50 points.** Optional network visualization: up to +10 bonus points. Partial credit may be awarded for incomplete or partially correct work. Numerical results without sufficient explanation or documentation may not receive full credit even when the reported values are correct.

**This is an individual assignment.** Each student must complete and submit their own work. Students should not collaborate on the analysis or submission.